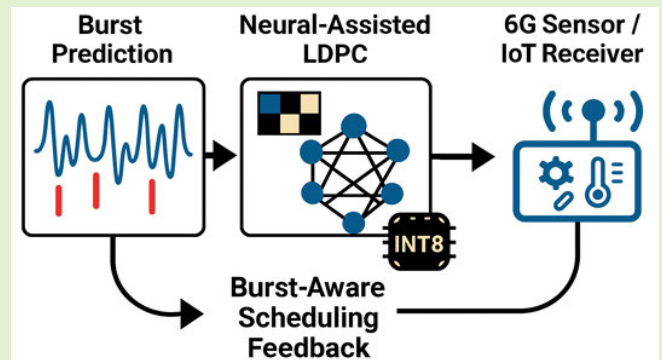


# Deep-Learning-Enabled and Quantization-Aware Burst-Resilient LDPC Decoding for 6G Sensor and IoT Communication Systems: A Review

Shiv Prasad Lingampally<sup>ID</sup>, Chakradhar Adupa<sup>ID</sup>, and Gyanendra Kumar Maurya<sup>ID</sup>

**Abstract**—Field measurements and recent sixth-generation (6G) mobile communication link-level studies consistently reveal that channel impairments manifest as correlated burst patterns rather than isolated symbol errors. Low-density parity-check (LDPC) codes continue to be the most widely adopted forward-error-correction scheme in these systems, yet experimental evidence indicates that conventional belief propagation (BP) decoders degrade once the burst length exceeds the code girth or lifting factor. Recent research has explored lightweight convolutional neural network (CNN) and recurrent neural network (RNN)-based models to detect regions affected by burst-induced reliability loss, and these indicators are used to adjust interleaver patterns or decoder scheduling. In hardware implementations, quantization-aware training (QAT), together with structured pruning, is used to limit numerical ranges and to lower computational complexity during field-programmable gate array (FPGA) and application-specific integrated circuit (ASIC) implementation. On the contrary to earlier surveys that addressed coding structures, learning-based decoders, or hardware quantization independently, this work brings together burst modeling, graph design, neural supervision, and quantization-aware deployment within a single unified framework. These observations are directly applicable to practical sensor networks, industrial Internet of Things (IoT) deployments, and edge receivers operating under burst errors and tight power constraints. This survey is designed to serve as a practical reference for engineers and researchers working on burst-resilient next-generation communication and sensing systems.

**Index Terms**—Burst-error resilience, deep-learning-assisted decoding, industrial Internet of Things (IoT), low-density parity-check (LDPC) codes, quantization-aware training (QAT), quasi-cyclic LDPC (QC-LDPC), sixth-generation (6G) wireless communication, spatially coupled LDPC (SC-LDPC), wireless sensor networks.



## I. INTRODUCTION

SIXTH-GENERATION (6G) wireless systems are designed to support ultra-reliable low-latency communication (URLLC), targeting block error rates below  $10^{-7}$  and end-to-end latency on the order of 1 ms for mission-critical services [1], [2]. However, these targets become difficult to maintain when burst-like disturbances occur. These burst patterns may arise due to factors such as multipath fading, mobility-related handovers, or interruptions linked to

nonterrestrial or UAV-assisted communication paths, which often degrade the link reliability in practical deployments [3].

This survey examines low-density parity-check (LDPC) decoding under burst impairments and how it interacts with learning-based supervisory modules and QAT-based hardware implementations. Broader topics such as Turbo/Polar decoding and generic ML architectures fall outside our scope. LDPC codes are widely recognized for achieving near-capacity performance over AWGN channels, owing to the sparse graph structures that enable efficient iterative message-passing decoding. When the channel produces long burst sequences, the decoder often gets stuck in trapping-like configurations, and the BLER may rise above  $10^{-5}$  even at moderate SNR. Because burst errors introduce temporal correlation, the standard memory-less-channel assumption no longer holds, leading to stalled updates in specific check-node groups and reduced belief propagation (BP) convergence [3], [4].

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The authors are with the Department of Electronics and Communication Engineering, School of Engineering, SR University, Warangal, Telangana 506371, India (e-mail: 2405c40039@sru.edu.in; chakradhar.a@sru.edu.in; gyanendrakumar.maurya@sru.edu.in).

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Previous surveys on 6G channel coding typically treat burst behavior only indirectly and rarely bring all related aspects together. LDPC-oriented surveys focus on structural design, AI-assisted decoding surveys emphasize neural improvements, and hardware surveys discuss quantization separately. None of them jointly analyzes how burst statistics influence LDPC structure, how learning modules supervise decoding, and how quantization constrains real deployments.

In most existing reviews, burst modeling, LDPC design, neural supervision, and hardware quantization are treated largely in isolation.

This separation obscures how burst behavior influences both decoding and hardware execution. Our aim is to bring these separate topics together in one place. Several recent works explore the use of learning-based components alongside the LDPC decoder. Some works apply CNN-based classifiers to detect reliability fluctuations in the LLR stream and highlight burst-affected symbol ranges. Some researchers have used RNN-based models to adjust the iteration schedule or damping settings, while others have applied RL methods to tune interleavers or lifting choices to avoid alignment between bursts and parity cycles [5], [6].

These learning modules assist BP by supplying auxiliary signals that influence either iteration count, damping, or interleaver selection and can help bypass harmful graph configurations without modifying the underlying LDPC code.

In parallel, several works concentrate on LDPC structures that inherently cope better with burst errors. Spatially coupled LDPC (SC-LDPC) codes enable a progressive, wave-like decoding behavior that incrementally mitigates burst-affected regions [4], whereas quasi-cyclic LDPC (QC-LDPC) codes exploit structured lifting patterns that facilitate burst mitigation and simplify hardware implementation [7]. Several works also study windowed decoding and low-power architectures for high-throughput operation to meet high-throughput hardware requirements [8]. On the hardware side, quantization-aware training (QAT), pruning, and mixed-precision updates (for example, 8-bit extrinsic and 4-bit intrinsic messages) help cut computation and memory use. With these techniques, many field-programmable gate array (FPGA)/application-specific integrated circuit (ASIC) designs can exceed 2 Gb/s while staying within practical power limits [8].

Here, we examine these approaches together and point out how they influence one another during design. We cover burst-channel models [such as the Gilbert–Elliott (GE)], LDPC code families (SC-LDPC, QC-LDPC, and protograph-based designs), deep-learning-enhanced decoding methods, and hardware-efficient quantization workflows. In this survey, performance comparison typically relies on the following measures.

1) *Burst Coverage Ratio*: The fraction of  $L_b$ -length bursts decoded successfully.

2) *Maximum Correctable Burst Length ( $B_{max}$ )*: The longest contiguous burst recoverable alongside latency ( $<1$  ms), throughput, and energy-per-bit constraints.

TABLE I  
COMPARISON OF PRIOR SURVEYS AND THEIR COVERAGE GAPS

| Survey                       | Focus                   | Gap                               |
|------------------------------|-------------------------|-----------------------------------|
| DL for channel coding (2024) | Neural decoders         | No burst modeling, no QAT         |
| LDPC surveys (2020–24)       | Code construction       | No neural or hardware integration |
| 6G coding overview (2024)    | High-level ECC          | No burst-specific metrics         |
| This survey                  | Burst + LDPC + AI + QAT | Unified co-design (first)         |

#### Summary of Contributions:

1) A unified taxonomy linking burst-channel models, LDPC structures, neural supervisory control, and QAT-based hardware design.

2) A critical comparison of SC-LDPC, QC-LDPC, and protograph LDPC under burst stress using burst coverage ratio (BCR),  $B_{max}$ , latency, and hardware metrics.

3) First consolidated review of learning-assisted LDPC decoding covering burst localization, iteration scheduling, lifting/interleaver optimization, and the reduced-precision deployment.

4) A cross-domain analysis of burst characteristics across PLC, FSO, NTN, and memory channels, revealing inconsistencies in evaluation practices.

5) We provide a codesign roadmap linking burst prediction, LDPC selection, decoding schedule, QAT deployment, and digital-twin validation.

Table I reviews the ways in which earlier surveys addressed LDPC design, burst-error modeling, learning-based decoding, and hardware aspects, and it highlights the gaps that motivate the present study.

Previous LDPC surveys typically address code construction or decoding, while DL-focused surveys discuss neural models without linking them to LDPC structure or decoding strategies, while deep-learning surveys analyze neural architectures without connecting them to LDPC structure. Hardware-focused works address quantization but ignore burst characteristics. Because these works address separate pieces, they do not provide a system-level perspective needed for 6G receivers, where burst statistics, graph structure, learning-based supervision, and fixed-point hardware behavior are tightly interdependent.

The surveyed literature consistently indicates that quantization-aware and learning-assisted LDPC decoding frameworks can improve burst resilience and hardware efficiency compared to conventional fixed-precision BP decoders. While the reported gains vary across channel models and implementation platforms, common benefits include reduced energy consumption, improved tolerance to extended burst spans, and lower average decoding latency under correlated error conditions.

TABLE II  
BURST-ERROR CHANNEL MODELS AND EVALUATION CONTEXTS USED IN LDPC-BASED STUDIES

| Channel / Error Model                  | Key Characteristics                                                            | Typical Use-Cases                                | Representative References |
|----------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------|---------------------------|
| Gilbert–Elliott (GE) Model             | Two-state Markov model capturing error correlation and burst persistence       | Wireless links with memory, IoT, sensor networks | [4], [7], [27], [36]      |
| Finite-State Markov Channel (FSMC)     | Multi-state extension of GE model enabling fine-grained burst severity control | Mobile and fading channels                       | [14], [27], [88]          |
| Magnetic / Storage Burst Model         | Long contiguous error bursts caused by media defects                           | Magnetic tape, NAND flash, data storage          | [5], [9], [30], [67]      |
| Free-Space Optical (FSO) Burst Channel | Atmospheric turbulence and pointing errors leading to burst losses             | Optical wireless and satellite links             | [15], [60], [130]         |
| Packet-Level Burst Erasure Model       | Consecutive packet losses due to congestion or blockage                        | Streaming, HARQ-based systems                    | [26], [84], [88]          |
| Memory-Affected Wireless Channel       | Temporal error correlation due to fading and interference                      | 6G sensor and IoT communications                 | [3], [27], [142]          |

The following sections build upon these foundations to systematically examine burst-error channel models, LDPC code structures, and representative learning-assisted decoding strategies, with an emphasis on their implications for practical and hardware-aware implementations.

## II. FOUNDATIONS

### A. 6G Constraints: Reliability, Latency, and Hardware Budgets

6G URLLC aims for a BLER below  $10^{-7}$  and submillisecond latency, even when the channel includes fading clusters, mobility-related disturbances, or NTN relays. Conventional BER/BLER metrics do not describe the cluster span or phase alignment; so additional burst-oriented metrics are required [16], [17], [18]. Hardware limits (5–10 pJ/bit, 1–5 Gb/s throughput) constrain decoder architectures on FPGA/ASIC platforms. In SC-LDPC sliding-window decoding (SWD), memory buffers scale with  $W \times N_v$  and without scheduling optimization latency can exceed 0.5 ms at 100 MHz [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58].

Therefore, proper evaluation should also consider: the BLER under structured burst insertion, 99th-percentile tail latency, and power and area footprints (LUT/FF/BRAM/DSP) [16], [17], [18].

Table II summarizes the dominant burst-error channel models adopted in the literature and highlights their key characteristics and typical application contexts. These models serve as the foundational abstractions for evaluating burst-resilient LDPC decoding strategies across wireless, storage, and optical communication systems.

Burst errors arise due to temporal correlation in channel impairments and are commonly modeled using simplified stochastic abstractions to enable tractable performance evaluation of channel codes. In the context of LDPC-based decoding, different burst-error channel models are adopted depending on the underlying physical medium, mobility conditions, and system-level constraints.

### B. Burst Error Models

Burst errors show time correlation, so the channel stops acting like an i.i.d. noise source. The GE Markov model

TABLE III  
BURST-ERROR CHANNEL MODELS AND THEIR DESIGN IMPLICATIONS

| Model Type       | Key Parameter(s) | Scenario            | Use in Design                                |
|------------------|------------------|---------------------|----------------------------------------------|
| GE Model         | $T_b$            | Fading clusters     | Interleaver tuning and burst span adjustment |
| SHMM             | $\alpha, \beta$  | Impulsive channels  | Damping scheduler and gating mechanisms      |
| Fading           | $M$              | High mobility       | SC window sizing                             |
| Phase-Coupled GE | $\Phi$           | Circulant alignment | Protograph lifting selection                 |

captures clustered “bad-state” sojourns; SHMM models represent impulsive interference in PLC/industrial settings; Fisher–Snedecor fading captures burst persistence in high mobility [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [151].

Table III summarizes how different burst models behave and outlines their effect on LDPC decoder design choices. The GE and SHMM models inform interleaver depth and damping control, while mobility-induced fading determines the sliding-window size in SC-LDPC decoding. Phase-coupled alignment indicates that QC-LDPC and protograph designs must avoid circulant-cycle matching, as such alignment can activate dominant trapping sets.

Fig. 1 shows an outline of the workflow used to build burst-tolerant LDPC decoders. Burst-channel models (e.g., GE, SHMM, and mobility-induced fading) describe the temporal structure of correlated error spans. These statistics inform LDPC code selection and construction (SC-LDPC, QC-LDPC, and protograph-based designs). When bursts align with QC-LDPC circulants, trapping sets can be activated; hence, the lifting pattern and interleaver settings become critical [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70]. These models guide practical choices such as window sizing, interleaver depth, and lifting parameters [23], [24], [25], [26],

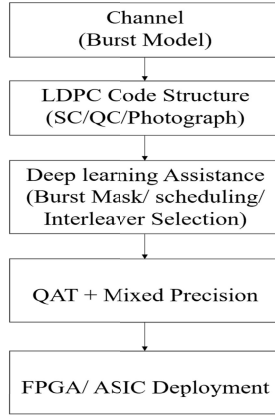


Fig. 1. Unified codesign pipeline illustrating how burst-channel statistics influence LDPC graph choices, neural supervisory modules, and quantization-aware hardware deployment.

[27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46].

Neural modules, such as CNN-based burst detectors, recurrent neural network (RNN) controllers, and RL-based interleaver or lifting tuners, modify how the decoder reacts when a burst occurs. Mixed-precision QAT enables stable fixed-point operation within tight FPGA/ASIC power and area limits.

### C. Burst-Aware Evaluation Metrics

Standard BER/BLER curves overlook parameters such as burst length and phase alignment [39], [40], [41], [42], [43], [44], [45], [46].

1) *Burst Coverage Ratio*: Fraction of bursts of length  $L_b$  corrected at target BLER.

2) *Maximum Correctable Burst Length ( $B_{\max}$ )*:

$$B_{\max} = \max \left\{ L_b : \text{BLER}(L_b) < 10^{-5}, \phi \in [0, 2\pi] \right\} \quad (1)$$

3) *Latency–Throughput Tradeoff*: Tracks the impact of window size  $W$  and iteration budget  $I$ .

4) *Energy per Bit*: QAT-enabled mixed precision preserves  $\geq 95\%$  BCR at 6-bit resolution while reducing power [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125].

5) In DT-based replay setups, identical burst traces are reused for consistent testing, while HIL systems verify that the decoder behavior matches hardware [140], [141], [142], [143], [144], [145].

## III. LDPC CODE FAMILIES

In 6G systems, the burst-handling ability of an LDPC code mainly depends on how its structure spreads error effects and reduces the chance of triggering trapping sets [7], [8], [9], [10]. We examine three widely deployed LDPC families—SC-LDPC, QC-LDPC, and protograph-based LDPC codes—with a focus on burst tolerance, latency characteristics, and practical hardware suitability [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58].

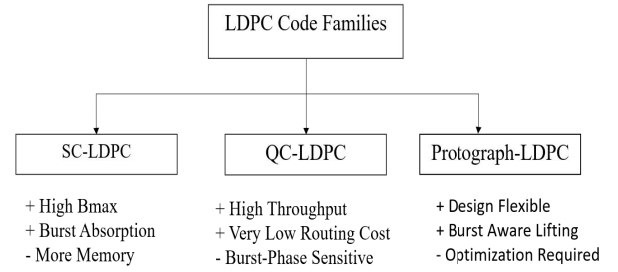


Fig. 2. Structural comparison of SC-LDPC, QC-LDPC, and protograph-based LDPC code families for burst-prone 6G systems. SC-LDPC supports wave-like decoding, QC-LDPC enables efficient hardware implementation, and protograph LDPC provides flexible graph construction with improved burst resilience.

### A. Spatially Coupled LDPC

SC-LDPC codes interconnect multiple sub-blocks through a band-diagonal parity-check structure, which enables a wave-style iterative decoding behavior [6], [16], [32]. When SWD with a window of size  $W$  is employed, the decoding wave progressively mitigates localized burst clusters, leading to strong burst-correction performance, with reported maximum correctable burst lengths  $B_{\max}$  of about 100–150 symbols and  $\text{BCR}(L_b)$  values above 0.90 for burst lengths up to 80 [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58]. Owing to the gradual propagation of reliability along the coupled chain, SC-LDPC codes maintain stable convergence even under correlated burst conditions [6], [16]. In practical hardware realizations, these advantages come at the cost of larger sliding-window buffers, as memory usage scales with  $W \times N_v$ , and the resulting decoding latency increases with the window size, typically reaching 0.5–0.9 ms at a 100-MHz operating frequency [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58].

Fig. 2 summarizes the structural differences among the LDPC families discussed here. SC-LDPC uses a coupled chain, which supports sliding-window operation and wave-style decoding. QC-LDPC uses circulant lifting to support structured memory access and efficient layered scheduling. Protograph-LDPC starts from a compact base graph that is expanded by lifting, which provides control over connectivity patterns and error-floor behavior.

### B. Quasi-Cyclic LDPC

QC-LDPC codes are constructed using circulant permutation matrices, allowing structured and predictable memory access while supporting efficient high-throughput layered decoding [17], [18], [19], [34], [35], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70]. For QC-LDPC implementations, Deng et al. [59] reported an FPGA throughput of 2.1 Gb/s using layered decoding, while Yang et al. [63] demonstrated a 4.5 Gb/s throughput on an ASIC-oriented architecture with optimized circulant scheduling. However, QC-LDPC codes generally correct shorter bursts than SC-LDPC codes, with  $\text{BCR}(L_b)$  around 0.70–0.80 and  $B_{\max}$  of roughly 60–80 symbols unless a burst-aware interleaver is used [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70]. QC-LDPC codes can be vulnerable to phase-aligned

bursts because they line up with circulant boundaries [36], [85]. Earlier work seldom analyzes how QC-LDPC behaves under phase-aligned bursts that activate circulant-based trapping sets.

### C. Protograph-Based LDPC

Protograph-based LDPC codes are derived from a compact base graph in which nodes and edges specify the local connectivity structure. After lifting by a factor  $Z$ , this structure expands into a full code while preserving the original graph relationships. This construction lets designers adjust degree profiles, cycle structure, and burst-sensitive regions through targeted lifting choices and controlled edge-spreading strategies, which helps improve resilience to long or phase-aligned bursts. Carefully designed protographs can achieve maximum correctable burst lengths  $B_{\max}$  of approximately 90–140 symbols, with  $\text{BCR}(L_b)$  values typically in the range of 0.85–0.92. This performance is close to SC-LDPC while avoiding the strict memory layout that QC-LDPC usually requires [49], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97]. However, the design process can be more involved because the lifting and edge-spreading choices significantly affect both burst tolerance and error-floor performance.

### D. Comparative Summary

The comparison assumes SNR values of 3–6 dB under a GE burst model. Hybrid MD-SC and SC-QC designs can improve  $B_{\max}$  by approximately 20%–25%, but they typically demand extra routing resources and buffer space [49], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97]. SC-LDPC benefits from the natural wave-like progression of its decoder, which often helps it recover from localized bursts, but it also needs larger buffers and extra control over the decoding window. QC-LDPC offers the most efficient hardware implementation; however, its structured circulant patterns may require burst-aware interleaving to prevent trapping-set activation.

The protograph-based LDPC provides greater flexibility to shape the local graph structure and improve robustness across different burst environments, although careful selection of lifting factors and edge-spreading patterns is required, as these significantly influence the resulting graph properties. Table IV presents a comparison of SC-LDPC, QC-LDPC, and protograph-based LDPC codes in terms of burst-correction capability, decoding latency, and hardware complexity. The results indicate that SC-LDPC offers stronger burst robustness, while QC-LDPC yields a more balanced compromise between error-correction performance and implementation efficiency.

## IV. DEEP-LEARNING-ASSISTED LDPC DECODING

In 6G LDPC decoding, burst errors usually show up as clusters of low-reliability LLR values, and this behavior limits the effectiveness of conventional BP. Prior research demonstrates that lightweight neural networks can monitor reliability variations in LLR patterns and guide certain BP update decisions during burst events, when bursts occur. Conventional BP

TABLE IV  
COMPARISON OF LDPC CODE FAMILIES UNDER  
BURST-PRONE 6G CONDITIONS

| Feature                       | SC-LDPC    | QC-LDPC                             | Protograph-LDPC   | Refs                 |
|-------------------------------|------------|-------------------------------------|-------------------|----------------------|
| BCR @ $L_b \leq 80$           | > 0.90     | 0.70–0.80                           | 0.85–0.92         | [47]–[97]            |
| $B_{\max}$ (symbols)          | 100–150    | 60–80                               | 90–140            | [47]–[97]            |
| Latency $\tau \approx 0.7$ ms | 0.5–0.9 ms | 0.3–0.6 ms                          | 0.5–1.0 ms        | [47]–[58], [59]–[70] |
| Hardware Complexity           | Medium     | Low (best)                          | Medium            | [47]–[97]            |
| Design Flexibility            | Medium     | Low                                 | High (best)       | [71]–[80]            |
| 6G URLLC Suitability          | Yes        | Yes (with burst-aware interleaving) | Yes (co-designed) | [47]–[97]            |

implicitly treats the incoming LLRs as if their reliability were fairly uniform across the Tanner graph, a condition that generally holds only under AWGN-type channels. During a burst, some parts of the LLR sequence drop noticeably, and this tends to activate familiar trapping-set patterns, increasing the error floor. Earlier studies introduced small auxiliary neural blocks around the BP loop to identify burst-prone regions, guide message updates, and adjust some structural choices using training data, while keeping the LDPC graph unchanged [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101].

### A. Burst Localization Networks

Burst errors cause localized changes in LLR amplitude and correlation. One limitation in earlier work is the absence of a common evaluation setup across different neural modules, LDPC variants, and quantization levels. Using GE or 3GPP channel traces, CNN- or Transformer-based models can learn to mark the symbols that lose reliability during a burst, producing a soft mask

$$\mathbf{M} \in [0, 1]^{N_v}. \quad (2)$$

Equation (2) represents the symbol indices exhibiting reduced reliability during a burst. The LDPC decoder applies selective damping and localized redecoding only in the masked regions, thereby avoiding additional iterations on parts of the codeword that have already converged. Prior studies report improvements in BCR and tail-latency behavior under burst-dominated channels [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [152].

### B. Adaptive Iteration Scheduling

Because the burst characteristics strongly influence when BP converges, a fixed iteration count often becomes inefficient—either too short or unnecessarily long. RNNs such as GRU or LSTM can monitor the syndrome sequence  $\mathbf{s}(t) = \mathbf{H}\mathbf{z}(t)$  along with the behavior of the LLRs to decide whether the decoder should continue or halt. These decisions reduce redundant

updates, cut the average iteration count, and improve consistency when bursts occur, especially in high-mobility fading environments [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118].

### C. Learning-Based Interleaver and Protograph Optimization

In QC-LDPC and protograph-based designs, burst-phase alignment with structured circulants can repeatedly activate harmful trapping sets. Reinforcement-learning-based and graph neural network (GNN)-based methods have been applied to explore lifting configurations and interleaver patterns that reduce harmful trapping-set alignments while preserving the underlying hardware structure. These methods improve  $B_{\max}$  and burst resilience while keeping the decoder complexity roughly unchanged [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136].

### D. QAT for Deployment

To meet 6G power and latency requirements, both neural modules and BP updates must operate under fixed-point arithmetic. In QAT, quantization is applied during training itself, allowing the neural blocks to run at 4–8-bit precision without a noticeable loss in decoding reliability. Mixed-precision arithmetic (e.g., 6-bit extrinsic, 4-bit intrinsic messages) has been shown to maintain decoding accuracy while keeping FPGA/ASIC power and silicon usage low. Existing surveys treat quantization as a standalone hardware issue, whereas in practice, it directly interacts with decoder convergence and burst recovery. Here, the interaction between quantization and decoder convergence is discussed in the context of reported QAT results for LDPC decoding, which are rarely analyzed jointly in earlier studies.

### E. Integrated Decoder Perspective

In real 6G receiver implementations, these neural units are placed next to the BP core and contribute to four key tasks. Convolutional neural network (CNN) or Transformer networks generate soft reliability masks that identify burst-affected regions and enable selective message updates [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101]. Recurrent units, such as GRU or LSTM regulate iteration flow based on the progression of syndrome values, thereby reducing latency and computational overhead [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118].

Fig. 3 shows the integrated neural-assisted LDPC decoding architecture. The CNN/Transformer module detects reliability degradation in burst-affected regions, guiding selective updates within the BP decoding core. The RNN-based scheduler dynamically adjusts the iteration flow to ensure timely convergence, while the RL-driven interleaver mitigates burst-cycle alignment. QAT combined with mixed-precision implementation enables energy-efficient and hardware-robust deployment on FPGA and ASIC platforms.

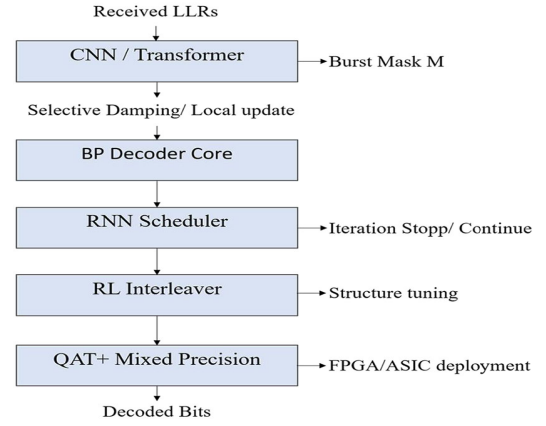


Fig. 3. Deep-learning-assisted LDPC decoding pipeline for burst-prone 6G systems. CNN/Transformer modules detect burst-affected regions, RNN units control iteration scheduling, RL-based interleavers reduce burst-cycle alignment, and QAT enables efficient FPGA/ASIC deployment.

Reinforcement-learning and graph-neural models optimize interleaver and lifting structures to minimize burst-cycle alignment and enhance trapping-set resilience [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136]. When QAT is combined with mixed precision, the decoder maintains stable performance even under low-bit hardware implementations such as FPGA and ASIC [45], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149]. Together, these components provide burst-awareness to the decoder while imposing only a small additional hardware cost. Their principal functions and reported performance improvements are summarized in Table V [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149].

Limitations of existing learning-assisted decoders, despite improved burst resilience, current DL-assisted LDPC designs suffer from four limitations.

- 1) Lack of unified training datasets across PLC/FSO/NTN channels.
- 2) Limited generalization to unseen burst distributions.
- 3) Absence of jointly optimized LDPC graphs for neural controllers.
- 4) Many works ignore quantization, leading to unrealistic power estimates.

#### Comparison to Traditional BP:

Traditional BP assumes uniform reliability and applies global updates, even when only localized regions of the codeword are affected. In these approaches, BP remains the main inference method, and the neural units simply add guidance signals that make the decoder more reliable under burst noise without adding much hardware cost [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108],

**TABLE V**  
NEURAL ASSISTED LDPC DECODING FUNCTIONS  
AND PERFORMANCE IMPLICATIONS

| Function                           | Neural Mechanism      | Decoder Interaction                     | Primary Benefit                                       | Refs        |
|------------------------------------|-----------------------|-----------------------------------------|-------------------------------------------------------|-------------|
| Burst region detection             | CNN / Transformer     | Selective update masking                | $\uparrow$ BCR,<br>$\uparrow$ $B_{\max}$              | [85]–[101]  |
| Iteration scheduling               | RNN / GRU             | Adaptive stop/continue policies         | $\downarrow$ Latency,<br>$\downarrow$ Iteration count | [102]–[118] |
| Interleaver / lifting optimization | RL / GNN              | Avoids burst-cycle structural alignment | $\uparrow$ Structural robustness                      | [119]–[136] |
| Deployment adaptation              | QAT + mixed precision | Fixed-point stable inference            | $\downarrow$ Energy/bit, FPGA/ASIC-ready              | [137]–[149] |

[109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149].

Table V depicts the neural-assisted modules integrated within the LDPC decoding pipeline. Each module contributes to a different stage of the decoding operation. CNN/Transformer models mark burst-affected regions, RNN/GRU units help determine how long decoding should continue, RL/GNN approaches improve lifting or interleaver choices, and QAT ensures that the neural parts operate reliably under low-precision hardware settings.

## V. PERFORMANCE ANALYSIS AND DESIGN IMPLICATIONS

Several factors influence how well an LDPC decoder handles burst errors, including the code's structure, the characteristics of the burst, and the way the decoder is controlled. The three major LDPC families considered—SC-LDPC, QC-LDPC, and protograph-based LDPC—behave differently when confronted with spatially correlated error clusters, affecting achievable burst coverage, latency, and hardware feasibility.

SC-LDPC codes distribute reliability gradually along the coupled chain, allowing the decoder to absorb localized burst errors more effectively, which progressively disperses localized error clusters across positions. When the burst is shorter than the window length, the decoding wave typically stabilizes the affected region without requiring structural modification. However, this property comes with increased buffering and scheduling overhead, which may affect latency in strict URLLC environments [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58].

QC-LDPC codes emphasize hardware regularity and throughput, benefiting from circulant-based parity-check matrices that support layered decoding and low routing complexity [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70]. However, the periodic circulant structure can create situations where certain burst positions repeatedly overlap with vulnerable cycle patterns, increasing decoding difficulty, where correlated errors reinforce absorbing sets. To mitigate this, burst-aware interleaving or circulant perturbation techniques are often necessary [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [85].

Protograph-based LDPC codes sit between the other two families in terms of design freedom and implementation cost. Through edge spreading and controlled lifting, designers can adjust the local cycle structure and redistribute burst-sensitive regions across the Tanner graph [71], [72], [73], [74], [75], [76], [77], [78], [79], [80]. This allows designers to improve burst resilience while keeping degree constraints and decoding complexity nearly unchanged. Neural modules act more like supporting controllers that work alongside BP, rather than replacing the decoding process itself. Burst-localization networks identify regions of reliability collapse and enable selective message adjustment [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101]. Recurrent controllers adjust iteration flow based on syndrome evolution to reduce unnecessary updates [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118]. Reinforcement-guided interleaver and lifting refinement reduces structural vulnerability to burst alignment [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136]. For practical deployment, these neural components must operate under QAT together with mixed-precision arithmetic to ensure fixed-point stability on FPGA/ASIC targets [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149].

This improves burst resilience without violating the throughput, energy, or latency constraints required in 6G edge receivers. In practice, burst tolerance reflects combined decisions in channel modeling, code design, decoding strategy, and hardware precision. When these elements are optimized separately, systems may underestimate burst vulnerability or fail to satisfy implementation constraints. A coordinated codesign approach is therefore necessary to maintain both reliability and efficiency under burst-dominated 6G conditions.

Table VI outlines the key performance tradeoffs among SC-LDPC, QC-LDPC, protograph-based, and neural-assisted LDPC decoders. SC-LDPC offers strong burst recovery with added latency, QC-LDPC achieves strong hardware efficiency but offers lower burst resilience, protograph designs enable structural flexibility, and neural-assisted decoding achieves balanced gains in reliability, latency, and energy efficiency.

The roadmap highlights the sequential dependencies between channel modeling, LDPC construction, neural supervision, hardware quantization, and system-level adaptation. Earlier stages focus on the accurate modeling of burst statistics, followed by the design of structurally robust LDPC graphs, then lightweight supervisory neural modules that guide

TABLE VI  
COMPARATIVE SUMMARY OF LDPC DECODING FRAMEWORKS IN BURST-PRONE 6G SYSTEMS

| Dimension         | Burst Coverage (BCR)                 | Maximum Burst Length                     | Latency Behavior                               | Hardware Complexity                   | Suitable Deployment              |
|-------------------|--------------------------------------|------------------------------------------|------------------------------------------------|---------------------------------------|----------------------------------|
| SC-LDPC           | High (wave propagation)              | Strong but window-size limited           | Higher due to window buffering                 | Medium                                | URLLC with moderate resources    |
| QC-LDPC           | Moderate (phase-sensitive)           | Limited without burst-aware interleaving | Low, supports high-throughput layered decoding | Low (best hardware efficiency)        | High-throughput FPGA/ASIC        |
| Protograph LDPC   | High (structure tunable via lifting) | Controllable through edge spreading      | Moderate latency                               | Medium                                | Flexible cross-channel co-design |
| DL-Augmented LDPC | High (burst mask + adaptive control) | Enhanced via selective supervision       | Reduced via RNN-based iteration scheduling     | Low/Medium with QAT + mixed precision | Edge AI/6G hardware co-design    |
| Refs              | [47]–[58]                            | [59]–[70], [71]–[80]                     | [47]–[58], [59]–[70]                           | [71]–[80], [85]–[136]                 | [137]–[149]                      |

decoding under correlated impairments. Precision-aware BP and QAT ensure hardware feasibility, while digital-twin feedback provides long-term adaptability. This roadmap arranges the main research directions into a workable sequence for the 2025–2030 period.

Table VII provides a consolidated failure-oriented view of LDPC decoding behavior in burst-dominated 6G channels. This analysis connects failure causes with structural, algorithmic, and hardware mitigation steps.

## VI. FUTURE RESEARCH DIRECTIONS

Although recent advances have improved the robustness of LDPC decoding under burst-error conditions, several challenges remain before these techniques can be deployed in practical 6G systems. These challenges span channel modeling, LDPC graph construction, neural supervisory integration, and hardware-efficient implementation [12], [13], [14], [15], [16], [17], [18], [19], [20], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80].

### A. Cross-Domain Burst Characterization

Most evaluations rely on simplified stochastic burst models, such as the GE process [12], [14], [18], [29], which do not completely reflect the characteristics of practical 6G environments, including PLC networks, FSO links, and high-mobility NTN channels [15], [19], [27], [31]. In practice, these systems show burst patterns that shift over time and across scales, shaped by interference levels and user mobility. Future work should particularly address the following points.

- 1) The unified burst-error representations across physical domains [12], [13], [14], [15], [16], [17], [18], [19], [20].
- 2) Standardized datasets for reproducible evaluation [27], [31].
- 3) Online burst-state estimation for real-time adaptation [18], [44].

Improved modeling will help close the gap between simulation behavior and real deployment conditions by providing better

burst statistics of decoder performance from simulation to deployment. A unified roadmap emerges from our analysis: 1) standardized burst datasets across wireless/PLC/NTN domains; 2) LDPC graph designs co-optimized with neural supervision; 3) QAT-driven precision-allocation frameworks; and 4) digital-twin-based deployment loops that enable real-time adaptation. Such a cross-layer roadmap is largely missing in earlier surveys, yet it is essential for practical 6G implementation.

### B. Integrated Codesign of Codes, Neural Control, and Hardware

When LDPC construction, neural add-ons, and quantization are handled independently, the resulting decoder usually fails to achieve a good balance among reliability, latency, and power use [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70].

To achieve better reliability–latency tradeoffs, upcoming decoder designs need closer coordination between code structure, neural control, and hardware precision [153], where:

- 1) LDPC graph topology and lifting patterns [71], [72], [73], [74], [75], [76], [77], [78], [79], [80];
- 2) message-passing schedules and damping strategies [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101];
- 3) bit-precision allocation across compute paths [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149] are optimized jointly for specific burst environments.

### C. Low-Complexity Neural Supervisory Models

Neural enhancements must remain compatible with edge hardware constraints. Lightweight supervisory approaches have shown promise, including:

- 1) CNNs on compressed LLR or syndrome domains [49], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101];
- 2) GRU-based iteration control with predictable computational cost [102], [103], [104], [105], [106], [107], [108],

TABLE VII  
FAILURE MODES OBSERVED IN LDPC DECODING UNDER BURST CHANNELS

| Failure Mode                                 | Root Cause                                           | Impact on LDPC Decoding                                              | Observed In                        | Mitigation Strategy                                                             |
|----------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------|
| Trapping-Set Activation                      | Burst-phase alignment with circulant boundaries      | Rapid BLER increase, error floor $>10^{-5}$ even at moderate SNR     | QC-LDPC, poorly lifted protographs | Burst-aware interleaving, circulant perturbation, RL-based lifting optimization |
| Wave-Stall in Sliding Window                 | Window size too small vs. burst span                 | Decoding wave fails to propagate; BCR drops by 20–40%                | SC-LDPC                            | Increase window size, adaptive windowing, syndrome-driven window shift          |
| Correlation Amplification                    | Long bad-state sojourns in GE/SHMM channels          | Localized LLR collapse, oscillatory BP updates                       | All LDPC families                  | Selective damping, CNN-based burst mask, hybrid BP-MS updates                   |
| Precision Collapse under QAT-free hardware   | Aggressive quantization ( $\leq 4$ -bit) without QAT | Incorrect sign updates, unstable convergence, higher iteration count | FPGA/ASIC implementations          | QAT, mixed-precision design (6–8 bits), retraining with quantization ranges     |
| Neural Overfitting to Synthetic Burst Models | Training only on GE/Markov datasets                  | Fails to generalize to PLC/FSO/NTN bursts; mask misclassification    | CNN/Transformer burst detectors    | Data augmentation, cross-domain training, digital twin-based dataset replay     |
| Scheduling Instability                       | RNN controller reacts to noisy syndrome trajectories | Premature stopping or unnecessary iterations; latency spikes         | RNN-based decoders                 | Stability constraints, gating mechanisms, training with noisy traces            |
| Structural Resonance in Protographs          | Poor edge spreading / lifting factor choice          | Burst aligns with short cycles $\rightarrow$ poor Bmax               | Protograph LDPC                    | GNN/RL-guided lifting, multi-type protographs, edge-spread diversity            |

[109], [110], [111], [112], [113], [114], [115], [116], [117], [118];

3) RL/GNN-guided interleaver or lifting adaptation [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136].

The main challenge is ensuring that these models adapt to changing channel conditions without exceeding runtime or memory limits.

#### D. Precision-Aware Quantization and Mixed-Precision Arithmetic

While QAT supports robust fixed-point inference, the optimal bit allocation across message updates, neural modules, and memory pathways is still not fully addressed [137], [138], [139], [140], [141], [142], [143], [144].

Promising directions include the following.

- 1) Layer-wise precision sensitivity analysis [138], [142].
- 2) Runtime-adaptive quantization based on channel state [143], [149].
- 3) Approximate arithmetic tuned to LDPC convergence dynamics [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149].

These techniques are important for reducing energy use while maintaining reliable decoding in 6G receivers.

#### E. Digital Twin-Driven Adaptive Decoding

Digital twin frameworks enable real-time feedback loops linking measured channel conditions to decoder behavior [150]. This enables environment-aware parameter tuning,

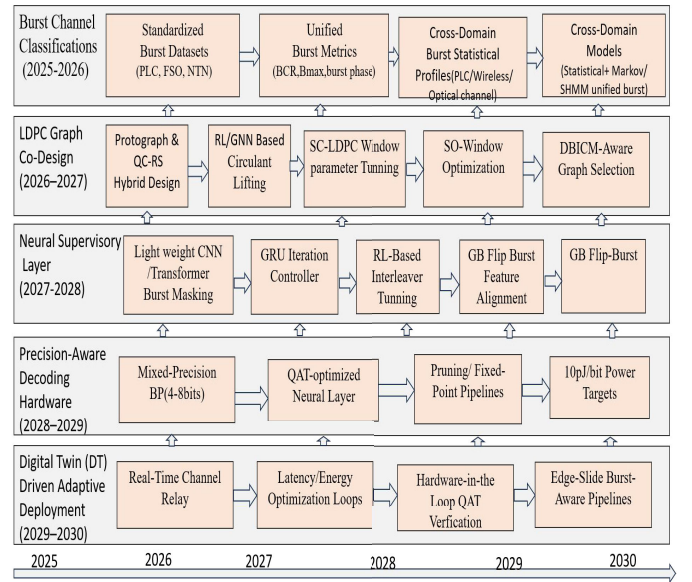


Fig. 4. Research roadmap for burst-resilient LDPC decoding in 6G systems over the 2025–2030 period. The roadmap shows the transition from burst-channel modeling and LDPC code design toward neural supervision, quantization-aware hardware implementation, and digital twin-enabled adaptive decoding.

adaptive scheduling under mobility, and stable performance under nonstationary burst patterns.

This feedback loop keeps testbed behavior aligned with the rapidly changing conditions seen in 6G links. Reliable burst-resilient LDPC decoding for 6G will require moving

from isolated algorithmic improvements toward a unified code-sign across channel modeling, LDPC structure, neural control, and hardware constraints [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133], [134], [135], [136], [137], [138], [139], [140], [141], [142], [143], [144], [145], [146], [147], [148], [149].

Fig. 4 illustrates a 2025–2030 research roadmap for burst-resilient LDPC decoder development, highlighting how the emphasis gradually moves from channel modeling and code design toward neural supervision, hardware-aware quantization, and digital twin-driven system adaptation.

## VII. CONCLUSION

This survey reviewed the key methods proposed to strengthen LDPC decoding under burst-error conditions, an issue of increasing importance in 6G systems where errors often occur in correlated clusters rather than independently. These burst patterns disrupt the iterative belief-propagation process and can lead to persistent error floors. This review shows that LDPC performance under burst conditions is strongly influenced by several interacting factors: the channel characteristics, the structure of the code, the update rules used in decoding, and the chosen hardware precision.

A comparison of spatially coupled, quasi-cyclic, and protograph-based LDPC families indicates that each family provides advantages under specific deployment requirements. Neural-assisted decoding adds adaptability by identifying burst-affected regions, adjusting iteration flow, and mitigating structural vulnerabilities in the code graph. With QAT and mixed-precision arithmetic, these methods can be deployed on FPGA/ASIC platforms with manageable hardware overhead.

We also found that improving only the code or only the decoder is not enough to handle burst errors effectively. Meaningful progress therefore depends on a coordinated design across channel modeling, LDPC construction, neural supervision, and hardware implementation. Future work should focus on establishing unified burst-channel datasets, developing lightweight neural controllers for edge devices, precision-aware message passing, and digital-twin-driven adaptive decoding. Coordinating these aspects is essential for maintaining reliable LDPC decoding across a range of 6G environments.

From an engineering point of view, these burst-resilient LDPC techniques provide a practical way to build low-latency and power-efficient 6G sensor and IoT receivers by combining QAT, mixed-precision processing, and lightweight neural support blocks. From a research point of view, future studies are expected to focus on common burst datasets, LDPC graph design guided by neural models, digital-twin-based adaptation, and better coordination between decoding methods and hardware limitations.

Overall, our analysis indicates the following five open research needs.

- 1) Standardized cross-domain burst datasets.
- 2) LDPC graphs co-designed with neural supervisors.
- 3) Precision-aware mixed-precision BP algorithms.
- 4) Runtime-adaptive decoders guided by digital twins.
- 5) Unified evaluation benchmarks for burst metrics (BCR/Bmax/latency).

These open issues shape a practical research path for future LDPC decoder designs. These findings directly support the design of field-deployable 6G sensors and industrial IoT receivers where burst noise, tight latency, and strict power budgets must be jointly addressed.

The findings reviewed in this work offer practical guidance for building reliable, low-power wireless sensing networks, enhancing smart-grid monitoring systems, and strengthening industrial IoT platforms that must operate under severe burst-error and interference conditions.

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**Shiv Prasad Lingampally** received the B.Tech. degree from Vardhaman College of Engineering, Shamshabad, Telangana, India, in 2014, and the M.Tech. degree from Kshatriya College of Engineering, Armoor, Nizamabad, Telangana, India, in 2018. He is currently pursuing the Ph.D. degree with the Department of Electronics and Communication Engineering, SR University, Warangal, India.

He has a strong academic background and is actively involved in research related to deep learning-assisted LDPC decoding, 6G wireless communication, burst-error correction, and hardware-aware optimization techniques.



**Chakradhar Adupa** received the Ph.D. degree in electronics and communication engineering from GITAM Deemed to be University, Visakhapatnam, Andhra Pradesh, India, in 2025, with specialization in VLSI design.

He is an Associate Dean of Innovation and Startups, the Director of NEST, and the Director of Entrepreneurship at the Science and Technology Centre, SR University, Warangal, India. With more than a decade of academic, research, and innovation experience, he has contributed to teaching, entrepreneurship development, and IEEE activities. His expertise includes VLSI design and verification, FPGA design, embedded systems, analog and digital electronics, and entrepreneurship development.

Dr. Adupa is actively involved with IEEE activities and has served in leadership roles, including Secretary of the IEEE CAS EDS Hyderabad Section. He has also organized and chaired multiple technical sessions, workshops, and entrepreneurship development programs.



**Gyanendra Kumar Maurya** received the Ph.D. degree in condensed matter physics from the Indian Institute of Information Technology, Allahabad (IIIT Allahabad), Prayagraj, India, in 2023, with a focus on topological insulators and optoelectronic devices.

He is an Assistant Professor of Electronics and Communication Engineering at SR University, Warangal, India. With over a decade of teaching and research experience, he has published more than 14 research papers. His expertise includes nanotechnology, semiconductor devices, and advanced material characterization for next-generation photodetectors.